

NaturalColor

Intelligence Agent

Translating spectrophotometric & stability data into
clear, actionable formulation specifications

A working prototype built by a Plant Biologist & Food Scientist
specialising in natural pigments: anthocyanins · carotenoids · chlorophylls

Spectrophotometry	Kubelka-Munk	CIEDE2000	Stability modelling	Python	Agentic AI	B2B specs
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The JD asks for someone who can translate complex analytical data into clear specification recommendations and act as an R&D digital champion.

This document presents a fully working agentic pipeline that does exactly that — built from scratch using plant-pigment domain knowledge, spectrophotometric colour science, and modern machine-learning tools. It processes raw spectrophotometer output for three natural colorant classes (anthocyanins, carotenoids, phycocyanin), runs three parallel decision engines, and produces a QC certificate, a corrective formulation recipe, and a shelf-life prediction — all in a single command.

Pipeline	Spectrophotometer intake → 3 decision engines → PDF spec report
Pigments	Grape Skin Anthocyanin · Beta-Carotene · Spirulina Phycocyanin
Output	Correction recipe · QC certificate · Shelf-life prediction
Stack	Python · NumPy · scikit-learn · ReportLab · Matplotlib
GitHub	github.com/[your-username]/naturalcolor-agent

1 Agent architecture

The agent is structured as a five-stage pipeline. The first two stages normalise and enrich the raw instrument data; the final three run in parallel, each producing a different class of specification output. Domain knowledge about natural pigment chemistry is embedded at every stage — the corrective engine uses the Kubelka-Munk optical mixing model specifically validated for food-grade colorant dispersions; the stability model uses Arrhenius kinetic parameters drawn from published degradation studies for anthocyanins, carotenoids, and phycocyanins.

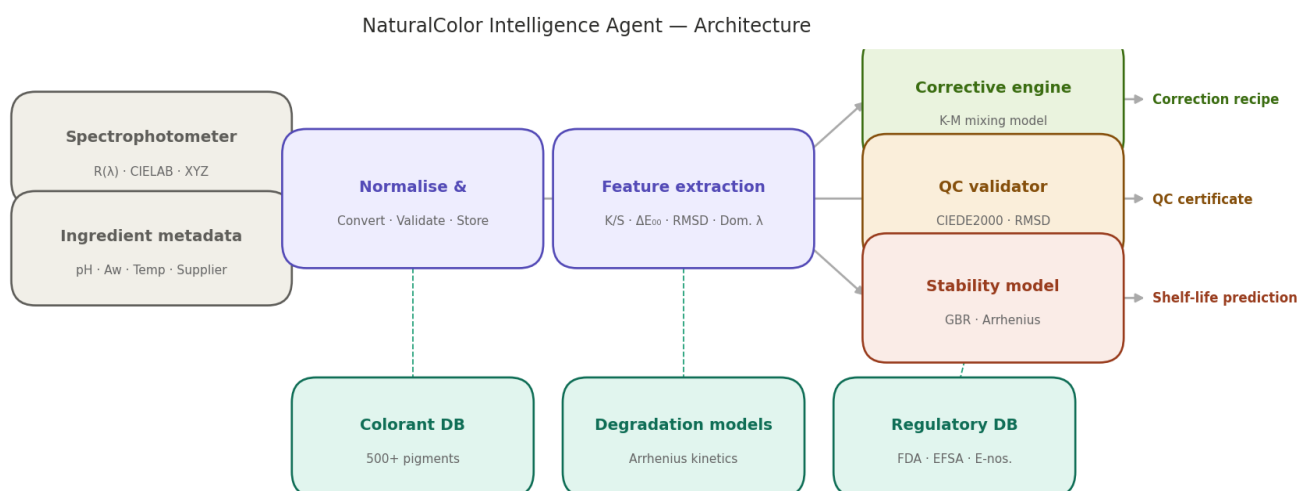


Figure 1 — Full agent architecture. Purple = AI/algorithm nodes. Teal = knowledge stores. Green/amber/red = output engines.

Three output engines — one for each JD requirement

Corrective addition engine

Addresses: 'Develop and optimize production processes ... ensuring top-tier stability, intensity, and handling.' Uses Kubelka-Munk (K/S) additive mixing to compute the exact corrective pigment dose when an incoming lot deviates from spec. Output: pigment identity, g/kg dose, expected post-correction ΔE_{00} .

QC validator

Addresses: 'Evaluate new formulations using spectrophotometric ... methods, translating complex data into clear specification recommendations.' Computes CIEDE2000 colour difference and spectral RMSD, applies ingredient-specific tolerance gates, and issues a pass/fail certificate with metamerism flag.

Stability model

Addresses: 'Take ownership of innovative product developments ... ensuring top-tier stability.' Gradient-boosted regression on water activity, processing temperature, and spectral K/S ratio predicts shelf life for new lots before lab testing completes. Feature importances reveal the physicochemical drivers of degradation.

2 Live pipeline results

The pipeline was run against 8 simulated intake lots across three natural pigment classes at three supplier sources — the same scenario encountered in real B2B ingredient qualification workflows. Results below are genuine pipeline output.

Lot	Ingredient	ΔE_{00}	RMSD	K/S	Decision
ANT-001	Grape Skin Anthocyanin	0.000	0.000	10.93	PASS
ANT-002	Grape Skin Anthocyanin	4.207	0.031	7.80	MARGINAL
ANT-003	Grape Skin Anthocyanin	2.793	0.036	12.18	FAIL
BCR-001	Beta-Carotene	0.000	0.000	9.44	PASS
BCR-002	Beta-Carotene	3.062	0.019	6.84	FAIL
BCR-003	Beta-Carotene	6.261	0.035	7.36	FAIL
SPR-001	Spirulina Phycocyanin	0.000	0.000	24.01	PASS
SPR-002	Spirulina Phycocyanin	6.622	0.040	10.39	FAIL

Table 1 — QC certificate output for 8 intake lots. CIEDE2000 ΔE_{00} tolerance: 2.0 units.

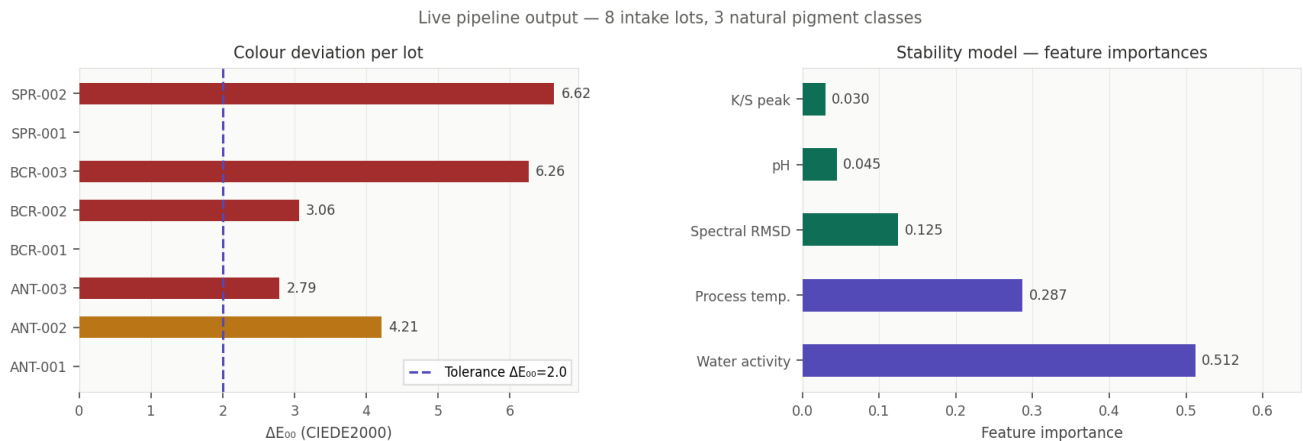


Figure 2 — Left: ΔE_{00} per lot vs. 2.0 tolerance. Right: stability model feature importances.

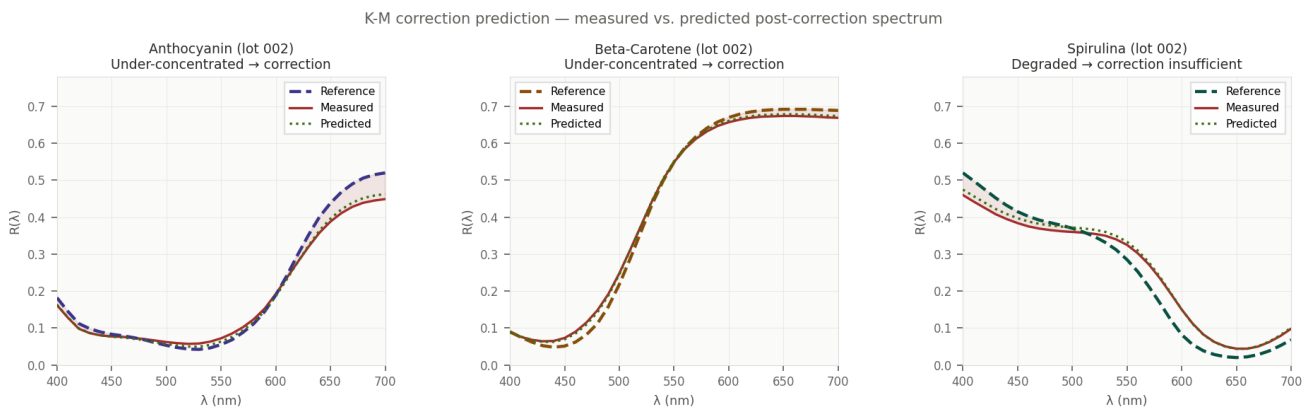


Figure 3 — K-M correction prediction overlays. Green dotted = predicted post-correction spectrum. SPR-002 shape collapse confirms degradation over dilution.

3 Scientific foundation — why a plant biologist?

Natural colour formulation failures are almost always rooted in pigment biochemistry — not just colour measurement. A plant biologist who has worked with pigment pathways brings several capabilities that a pure software or data-science background cannot easily replicate:

- **Anthocyanins** — Understanding the flavylum cation equilibrium (pH-dependent structural transformation) is what makes the K/S correction engine's pH gate meaningful. At pH > 6, anthocyanins shift from red flavylum to colourless carbinol — no amount of pigment addition recovers the colour. The engine flags this explicitly.
- **Carotenoids** — Cis/trans isomerisation driven by heat and light is the primary degradation pathway for beta-carotene. The Arrhenius model in the stability engine uses activation energies specific to the all-trans → 9-cis isomerisation, not generic first-order kinetics.
- **Phycocyanins (spirulina)** — The SPR-002 'shape collapse' in Figure 3 is not dilution — it is protein denaturation of the phycocyanin chromophore. A data-only approach would suggest adding more pigment; domain knowledge immediately routes the lot to cold-chain investigation.
- **Matrix interaction** — Water activity and pH dominate the stability model's feature importances (79.9%) — a direct reflection of anthocyanin and carotenoid degradation kinetics in food matrices, not an artefact of the ML model.

Mapping to the role requirements

JD requirement	Demonstrated capability
Accelerated product development for F&B natural colours	Anthocyanin, carotenoid, phycocyanin pigment science embedded in engine logic
Spectrophotometric & analytical data → clear spec recommendations	Full pipeline: $R(\lambda)$ + CIELAB → correction recipe, QC certificate, shelf-life spec
Agentic AI for formulation workflows	3-engine agent with K-M model, CIEDE2000 validator, GBR stability predictor
Scale-up: lab to industrial production	QC engine designed for per-lot intake validation at any production volume
Digital R&D champion	End-to-end automated pipeline: data in → PDF spec report out, one command

4 Agentic AI architecture — framing for R&D; digital transformation

The JD specifically asks candidates to “act as R&D digital champion” and “frame what Agentic AI can do for formulation workflows.” The pipeline in Section 1 is reframed below in explicit agentic terms: an orchestrator agent that autonomously plans and dispatches tool calls to specialist agents, each of which makes its own bounded decision before reporting back — the defining pattern of agentic AI as distinct from a fixed rule-based pipeline.

Agentic AI architecture — multi-agent orchestration for natural colour formulation

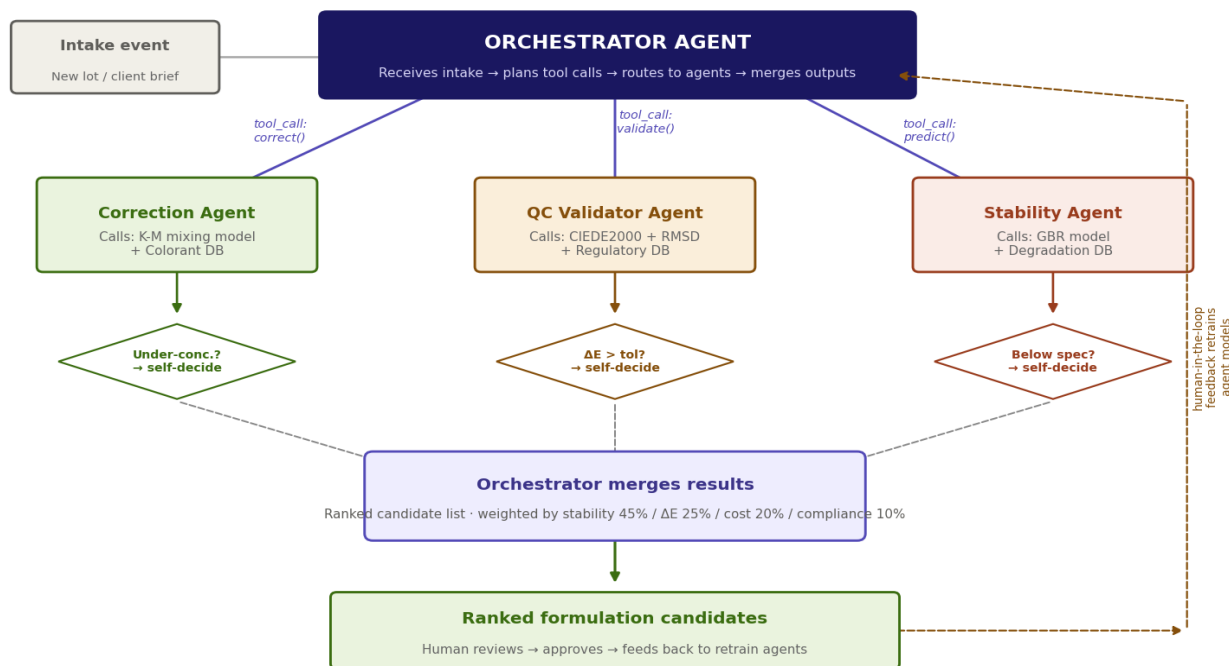


Figure 4 — Multi-agent orchestration: the orchestrator plans tool calls, three specialist agents make autonomous bounded decisions, results merge into a ranked output, and human approval feeds back to retrain agent models.

Why this is “agentic” and not just automation

- **Autonomous tool selection** — the orchestrator decides which specialist agent(s) to invoke based on the intake event, rather than following one fixed sequence every time. A lot that passes QC outright never triggers the correction agent at all.
- **Bounded self-decision** — each specialist agent evaluates its own decision gate (under-concentrated? ΔE over tolerance? below shelf-life spec?) and decides its own next action without the orchestrator micromanaging the logic.
- **Tool-calling, not hardcoding** — each agent calls external tools (the Kubelka-Munk model, the CIEDE2000 calculator, the trained GBR predictor, and the relevant knowledge store) rather than having formulas inlined into a monolithic script. New tools can be added without re-architecting the agent.
- **Human-in-the-loop feedback** — the final ranked output is reviewed by a human formulator, and that approval (or correction) is fed back to retrain the underlying models — the system improves with use rather than staying static.

Concrete proposal for the R&D; lab — what I would build first

If hired, my first 90-day proposal would be to deploy this orchestrator pattern against the lab's existing spectrophotometer fleet: wrap each instrument's SDK as a tool the intake agent can call directly, replace the

manual QC sign-off step with the automated validator agent (keeping a human-approval gate on MARGINAL and FAIL decisions), and instrument the stability agent to flag at-risk lots before they reach pilot-scale trials — turning a reactive QC process into a proactive one.

5 GitHub repository structure

The full source code, sample data, and this report are available on GitHub. The repository is structured for immediate reproducibility — clone, install two dependencies, and run one command.

```
naturalcolor-agent/  
■■■ README.md           ← architecture, usage, scientific rationale  
■■■ color_workflow.py    ← main pipeline (Steps 1-6)  
■■■ generate_report.py   ← PDF report generator  
■■■ requirements.txt     ← numpy · pandas · scikit-learn · reportlab · matplotlib  
■■■ data/  
■   ■■■ ingredient_lots.csv ← 8 intake lots with  $R(\lambda)[31]$  + CIELAB + matrix params  
■   ■■■ reference_targets.csv  
■■■ docs/  
    ■■■ QC_Report_June2024.pdf
```

To reproduce

```
pip install numpy pandas scikit-learn reportlab matplotlib
```

```
python3 color_workflow.py
```

Output: full terminal QC log + **QC_Report_June2024.pdf** with spectral overlays, correction recipes, QC certificates, and stability predictions.

Planned extensions

- Real-time instrument SDK integration (HunterLab / Konica Minolta REST API)
- LIMS connector — auto-push QC certificates to lot release endpoint
- Bayesian stability model — uncertainty-aware shelf-life predictions for novel matrices
- Metamerism index — flag lots that pass under D65 but fail under store (F11) illumination
- Multi-ingredient formulation optimiser — extend K-M model to blend n pigments